A Godunov-Type Finite-Volume Model for the Transient Blowdown of a Pressurized Vertical Well with Conjugate Completion–Formation Heat Transfer: Formulation, Verification, and Baseline Nitrogen Results

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Abstract

This document describes the formulation, numerical implementation, and verification of a one-dimensional transient compressible-flow model for the blowdown of a pressurized vertical well through a wellhead restriction. The governing mass, momentum, and total-energy equations are solved in conservative form with a Godunov-type finite-volume method employing MUSCL–Hancock reconstruction and the HLLC approximate Riemann solver, a numerical architecture selected on the basis of systematic shock-tube benchmarks against the split-coefficient-matrix method (SCMM) used in earlier work (Gessner, 2010). Gravity, wall friction, and fluid–wall heat transfer enter as source terms; the gravitational work term is discretized in an energy-consistent face-flux form that closes the discrete total-plus-potential energy budget to machine precision. The wellhead is modeled as a choke coupled to the interior solution through the outgoing characteristic, yielding a physically consistent choked/unchoked/closed discharge sequence. Verification comprises exact shock-tube solutions, conservation audits, the quasi-static lumped-vessel limit, and grid, time-step, and boundary-condition sensitivity studies. Baseline results are presented for the adiabatic blowdown of a 2000 ft nitrogen-filled well from 13.79 MPa. The model is then extended with a conjugate completion–formation heat-transfer formulation: at every axial cell, an implicitly advanced radial thermal network represents the concentric completion of the LSU CO_2 research well — 7-5/8 in CRA inner casing, air-filled annulus (natural convection and radiation), 13-3/8 in and 20 in cemented casings — and the surrounding formation with an undisturbed geothermal far field. The network is verified against analytical limits and an independent reference solution for transient radial conduction. For the baseline slow blowdown of the actual well geometry, heat ingress from the completion and formation nearly suppresses the expansion cooling (minimum wellhead temperature 284 K against 124 K for the adiabatic wall), with the air annulus identified as the dominant solid-side thermal resistance. The thermodynamic-property model is isolated behind an interface so that real-fluid CO_2 properties can replace the ideal-gas closure without modification of the flow solver.

1 Introduction

Transient depressurization (blowdown) of wells and pipelines containing compressed gas is a canonical problem in well control, CO_2 transport and storage operations, and the repurposing of oil and

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gas wells for energy storage. The rapid opening of a wellhead valve launches an expansion wave into the wellbore, followed by a long quasi-static discharge during which the contained fluid cools substantially by expansion work. For CO_2, the associated temperature excursions can cross the saturation locus, with consequences for well integrity and metering; a predictive transient model is therefore a prerequisite for the design and interpretation of blowdown and injection tests.

The present work is the third phase of a numerical framework whose starting point was the Fortran implementation of the split coefficient matrix method (SCMM) developed in Gessner (2010) following Chakravarthy et al. (1980) and Städtke (2006), with linear systems solved by the banded routine of Aziz and Settari (1979). In Phases 1–2 the original single-phase SCMM solvers (conservative and non-conservative formulations, first- and second-order upwind variants) were reconstructed in Python and validated against exact shock-tube solutions, reproducing the original error tables; Godunov-type finite-volume (FV) alternatives were then evaluated on identical benchmarks. The FV method with MUSCL–Hancock reconstruction (van Leer, 1984; Toro, 2009) and the HLLC approximate Riemann solver (Toro et al., 1994) proved approximately one order of magnitude more accurate than the best original scheme at two orders of magnitude lower computational cost, with exact discrete conservation — properties the quasi-linear SCMM discretization cannot provide. That evidence, rather than a priori preference, fixed the architecture adopted here.

This document records (i) the mathematical model and its source terms, (ii) the numerical method, including the wellhead choke boundary condition and an energy-consistent treatment of gravitational work, (iii) the verification campaign, and (iv) baseline results for adiabatic nitrogen blowdown of a 2000 ft well. Planned extensions — real-fluid CO_2 thermodynamics, and conjugate heat transfer with the completion and formation — are outlined in section 8.

2 Problem formulation

A vertical well of depth $L = 609.6\,\mathrm{m}$ (2000 ft) and constant circular cross-section (inner diameter D, flow area $A = \pi D^2/4$) is initially closed and filled with quiescent gas at uniform pressure p_0 and temperature T_0. The axial coordinate z is measured upward from the closed bottom ($z = 0$) to the wellhead ($z = L$). At $t = 0$ a choke of bore diameter d_c and discharge coefficient C_d at the wellhead is opened instantaneously to an external environment at back pressure p_b. The baseline configuration is summarized in table 1.

Table 1: Baseline case definition.

Fluid	N_2, ideal gas ($R = 296.8\,\mathrm{J/(kg\,K)}$, $\gamma = 1.40$)
Well depth L	609.6 m (2000 ft)
Well inner diameter D	0.100 m
Initial state	uniform, $p_0 = 13.79\,\mathrm{MPa}$ (2000 psi), $T_0 = 300\,\mathrm{K}$, $u = 0$
Choke	$d_c = 12.7\,\mathrm{mm}$, $C_d = 0.84$, $p_b = 101.325\,\mathrm{kPa}$
Bottom boundary	closed (solid wall)
Walls	adiabatic (baseline); friction neglected (baseline)

The quantities of interest are the axial–temporal fields $p(z,t)$, $T(z,t)$, $\rho(z,t)$, $u(z,t)$, the wellhead mass flow rate $\dot{m}(t)$, and the mass inventory $M(t)$.

3 Mathematical model

3.1 Governing equations

One-dimensional transient compressible flow along the well axis is governed by the conservation of mass, momentum, and total energy,

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u)}{\partial z} = 0, \tag{1}$$

$$\frac{\partial (\rho u)}{\partial t} + \frac{\partial}{\partial z}\left(\rho u^2 + p\right) = -\rho g \; - \; \frac{f_D \, \rho \, u \, |u|}{2D}, \tag{2}$$

$$\frac{\partial E}{\partial t} + \frac{\partial}{\partial z}\left[u \, (E + p)\right] = -\rho u g \; + \; \frac{4 \, q''_w}{D}, \tag{3}$$

where ρ is the density, u the velocity (positive upward), p the pressure, $E = \rho \, (e + u^2/2)$ the total energy per unit volume, e the specific internal energy, g the gravitational acceleration, f_D the Darcy friction factor, and q''_w the wall heat flux into the fluid. Wall shear appears only in the momentum equation: a stationary wall performs no net work on the fluid, and frictional dissipation is recovered automatically by the conservative total-energy formulation. The friction factor, when enabled, is evaluated from the all-regime correlation of Churchill (1977).

3.2 Equation of state

The system is closed by an equation of state supplying $(p, T, a) = f(\rho, e)$, where a is the speed of sound. The solver and the boundary conditions interact with the fluid exclusively through an abstract property interface comprising this closure, the isentrope $\rho(p;\, p_{\mathrm{ref}}, \rho_{\mathrm{ref}})$, the outgoing-wave (Riemann-invariant) velocity relation, and the isentropic-nozzle mass flux. For the present stage the interface is implemented by a calorically perfect gas,

$$p = \rho R T, \qquad e = c_v T, \qquad a = \sqrt{\gamma R T}, \tag{4}$$

with nitrogen properties; a real-fluid CO_2 implementation of the same interface (evaluating the isentrope and invariant numerically) replaces this class without modification of the flow solver.

3.3 Thermal boundary hierarchy

The wall heat flux q''_w is supplied by an interchangeable thermal model: (1) adiabatic wall, $q''_w = 0$ (fluid-dynamic baseline); (2) prescribed wall temperature with convective coupling, $q''_w = h \, (T_w - T)$, h from the Dittus–Boelter correlation (simplified benchmark); (3) prescribed or externally computed heat flux; and (4) the target formulation, conjugate coupling to transient radial conduction through the completion layers and the formation, with $q'' = -k_f \, (\partial T_f / \partial r)\big|_{r_w}$ emerging from the coupled solution. Level (4) is formulated and verified in section 7 for the completion of the LSU CO_2 research well.

4 Numerical method

4.1 Finite-volume discretization

Equations (1)–(3) are discretized on N uniform cells of width $\Delta z = L/N$ in conservative finite-volume form,

$$\mathbf{V}_j^{n+1} = \mathbf{V}_j^n - \frac{\Delta t}{\Delta z}\left(\mathbf{F}_{j+1/2} - \mathbf{F}_{j-1/2}\right) + \Delta t \, \mathbf{S}_j, \qquad \mathbf{V} = \left[\rho, \; \rho u, \; E\right]^T, \tag{5}$$

with interface fluxes from the HLLC approximate Riemann solver (Toro et al., 1994; Toro, 2009) acting on MUSCL–Hancock-reconstructed primitive variables (van Leer limiter). The HLLC solver restores the contact wave omitted by HLL, which is immaterial for the smooth blowdown interior but retained for consistency with the shock-capturing benchmarks and for robustness in later two-phase extensions. Time integration is explicit under the CFL condition $\Delta t = \mathrm{CFL}\,\Delta z / \max_j (|u_j| + a_j)$, with $\mathrm{CFL} = 0.8$ unless stated otherwise.

4.2 Source-term treatment

Gravity is included in the Hancock predictor half-step (maintaining formal second order), and the momentum source $-\rho g$, friction, and wall heat flux are advanced with a Heun (trapezoidal) corrector. The gravitational *work* term in eq. (3) requires care: evaluating it as $-\rho_j u_j g$ at cell centers leaves a residual in the discrete energy budget that does *not* converge with mesh or time-step refinement, because the potential-energy inventory evolves according to the *face* mass fluxes while the work term samples cell-centered momenta. The term is therefore discretized in the energy-consistent face-flux form

$$(-\rho u g)_j \;\longrightarrow\; -\,g\,\frac{F^{\rho}_{j-1/2} + F^{\rho}_{j+1/2}}{2}, \tag{6}$$

where F^{ρ} denotes the time-centered HLLC mass flux. With eq. (6) the discrete budget

$$\sum_j \left(E_j + \rho_j g z_j\right) A\,\Delta z \;+\; \int_0^t \dot{m}\,\left(h_0 + gL\right)\,dt' \;=\; \text{const} \tag{7}$$

telescopes exactly (machine precision), where h_0 is the stagnation enthalpy of the effluent. The scheme is not well-balanced in the sense of preserving discrete hydrostatic equilibria exactly; residual velocities of order 5 cm/s (200 cells) persist in a closed hydrostatic column, which is negligible against blowdown velocities but should be revisited if long-time post-blowdown equilibria become of interest.

4.3 Bottom boundary

The closed bottom is an exact solid wall: the wall-face left state is the mirror image (in velocity) of the MUSCL-reconstructed interior right state, for which the HLLC mass and energy fluxes vanish identically.

4.4 Wellhead choke boundary condition

The wellhead must not be treated as a fixed-pressure boundary while a restriction separates the well from the environment. The pipe-side face state (ρ_f, u_f, p_f) at $z = L$ is instead required to satisfy, simultaneously:

1. *Coupling to the interior.* The face lies on the isentrope through the adjacent interior state (ρ_i, u_i, p_i) and on its outgoing C^+ characteristic,

$$u_f = u_i + \frac{2}{\gamma - 1}\left(a_i - a_f\right), \qquad \rho_f = \rho_i\left(p_f / p_i\right)^{1/\gamma}; \tag{8}$$

2. *Valve mass balance.* The flow accepted by the choke, modeled as a quasi-steady isentropic nozzle fed by the stagnation state (p_{0f}, T_{0f}) of the face flow, equals the pipe efflux:

$$\rho_f u_f A = C_d A_t \, G(p_{0f}, T_{0f}, p_b)\,, \qquad A_t = \pi d_c^2/4, \tag{9}$$

where G is the standard compressible-orifice mass flux, which is choked when $p_b/p_{0f} \le (2/(\gamma+1))^{\gamma/(\gamma-1)}$ and subcritical otherwise.

Substitution of eq. (8) into eq. (9) yields a scalar equation for p_f, solved by Brent iteration on a bracket bounded below by the *sonic face pressure* — the C^+ coupling forbids a supersonic pipe exit; if the valve demands more than the sonic pipe supply, the exit itself runs sonic, recovering the open-pipe choking limit $A_t \to A$ — and above by the pressure at which eq. (8) brings the face to rest. When the interior stagnation pressure falls below p_b, the valve closes (check-valve clamp; backflow excluded). The resulting face flux $\left[\rho_f u_f,\ \rho_f u_f^2 + p_f,\ u_f(E_f + p_f)\right]$ replaces the HLLC flux at the top face, so the interior update and the discharge remain mutually consistent through the entire choked → unchoked → closed sequence as the well depressurizes, and $\dot{m}(t) = \rho_f u_f A$ is obtained as part of the solution.

5 Verification

5.1 Shock-capturing heritage

The HLLC/MUSCL–Hancock kernel was verified in Phase 2 on the single-phase shock tube of Gessner (2010) (air; 500 kPa, 400 K / 100 kPa, 300 K; 4001-point grid) against the exact Riemann solution. Table 2 summarizes the mean absolute errors: the first-order Godunov scheme already matches the best legacy SCMM variant, and MUSCL–Hancock improves on it by an order of magnitude at a small fraction of the cost, with discrete conservation to machine precision. The HLLC flux is indistinguishable from the exact Riemann flux on this problem.

Table 2: Single-phase shock-tube benchmark at $t = 3$ ms: mean absolute errors against the exact solution (4001 nodes / 4000 cells). SCMM rows: Python reconstruction of Gessner (2010); thesis values in parentheses.

Scheme	p [Pa]	u [m/s]	ρ [kg/m^3]	T [K]	Wall time [s]
SCMM non-cons., 2nd order	3027 (2713)	4.29	0.0230	2.02	186
SCMM conservative, 2nd order	383 (312)	0.44	0.0041	0.35	399
FV first order, HLLC	316	0.28	0.0041	0.41	1.4
FV MUSCL–Hancock, HLLC	34.9	0.040	0.00057	0.065	2.0

5.2 Conservation audits

For the closed well the solution remains identically at rest. For the discharging well with gravity active, the mass balance $M(t) + \int_0^t \dot{m}\, dt' = M(0)$ and the energy balance eq. (7) close to $\mathcal{O}(10^{-15})$ relative error in all cases reported here, including the open-pipe limit. During development, the cell-centered gravitational work term was found to leave a non-convergent budget residual of $\mathcal{O}(10^{-4})$; the face-flux form eq. (6) removes it identically (section 4.2).

5.3 Quasi-static lumped-vessel limit

With gravity deactivated, the 1-D solution must approach the zero-dimensional adiabatic-vessel model (isentropic residual gas, quasi-steady orifice discharge) whenever the acoustic transit time $L/a_0 \approx 1.7\,\mathrm{s}$ is small against the discharge time scale ($\approx 200\,\mathrm{s}$ for the baseline choke). Figure 1 shows wellhead pressure, mass flow rate, and inventory against the ODE reference: maximum relative deviations are 0.9 % (p) and 0.8 % ($\dot{m}$) after the initial wave transient. With larger chokes the *wellhead* pressure sits several percent below the column mean — a genuine 1-D effect of the sustained expansion toward the choke — while the mass inventory continues to track the lumped model to 0.5 %.

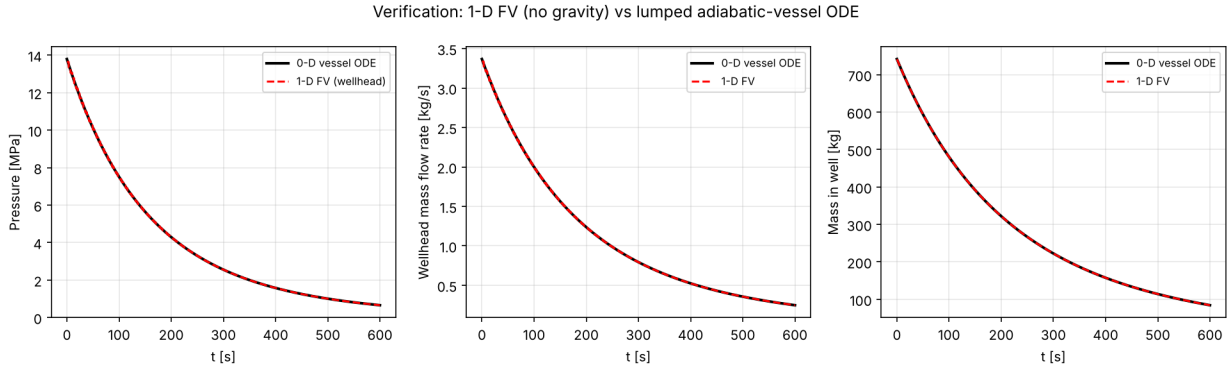


Figure 1: Verification against the lumped adiabatic-vessel ODE (gravity off): wellhead pressure, mass flow rate, and mass inventory.

5.4 Grid, time-step, and boundary sensitivity

For the baseline case over the first 60 s, the maximum relative deviation of $\dot{m}(t)$ from an 800-cell reference is 2.6 % (100 cells), 0.88 % (200), and 0.26 % (400); reducing the CFL number from 0.9 to 0.5 and 0.2 changes $\dot{m}(t)$ by at most 0.6 % and 1.2 %, respectively. A choke-bore sweep from 1/4 in to the open-pipe limit (fig. 2) yields initial mass flow rates scaling with throat area as expected (0.84/3.32/12.9/46.6 kg/s, open pipe 143 kg/s), monotonic throughout, stable in the pipe-choked limit, and conservative to machine precision in every case.

6 Baseline results: adiabatic nitrogen blowdown

6.1 Wave propagation

Figure 3 presents the distance–time diagram of the pressure perturbation $p - p_{\mathrm{init}}(z)$ during the first 8 s, for a run initialized in isothermal hydrostatic equilibrium so that the valve-opening wave is the only transient (a uniform initial column superposes a gravitational settling wave of comparable amplitude). Opening the choke launches an expansion wave that descends the well at the sound speed, reaching the bottom at $t = L/a_0 \approx 1.7\,\mathrm{s}$, reflecting, and returning to the wellhead at $2L/a_0 \approx 3.4\,\mathrm{s}$; since the choke passes only a small fraction of the sonic pipe flux, the wellhead acts as a nearly closed end and the wave re-reflects as an expansion, producing the staircase depressurization of period $2L/a_0$ with steps of $\mathcal{O}(130\,\mathrm{kPa})$ visible in the figure. Animations of the full field evolution accompany this document. The initial expansion is weak relative to the inventory

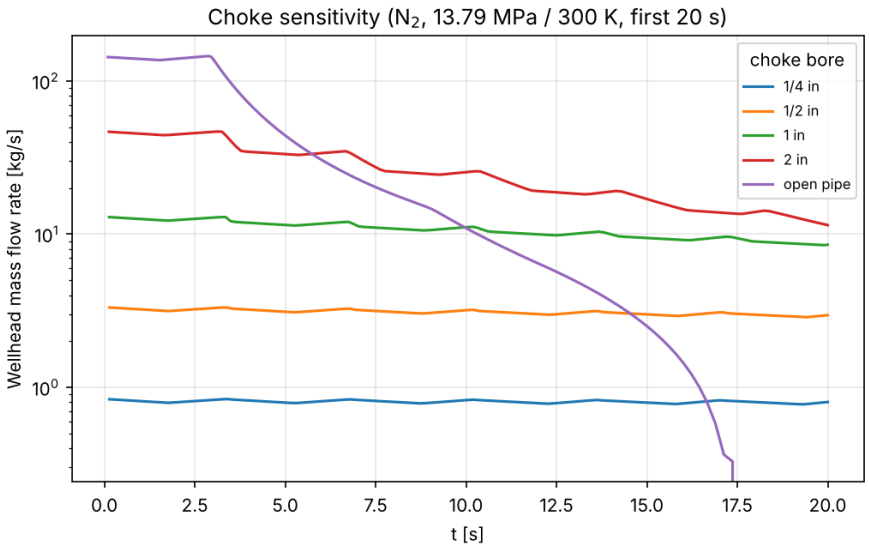


Figure 2: Sensitivity of the wellhead mass flow rate to the choke bore, including the open-pipe limit.

pressure ($\rho a\,\Delta u \sim 1.5 \times 10^5$ Pa against 13.79 MPa), and the subsequent discharge is quasi-static — consistent with the lumped-vessel agreement above. Larger chokes proportionally strengthen the initial wave.

6.2 Depressurization and cooling

Figures 4 and 5 show the field profiles and wellhead histories over 600 s. The column depressurizes nearly uniformly (mild top-to-bottom gradient sustained by the discharge plus the hydrostatic head), the velocity profile is linear in z — the signature of spatially uniform expansion — with superposed decaying acoustic oscillations, and the discharge remains choked for the entire interval shown. Expansion work cools the inventory continuously; by $t = 600$ s the wellhead fluid reaches 127 K, i.e. essentially the critical temperature of nitrogen (126.2 K).

Two implications follow. First, the ideal-gas closure is increasingly strained precisely where the blowdown terminates, quantifying the need for the real-fluid property stage. Second, CO_2 under comparable conditions would intersect its saturation locus far earlier in the transient, so phase appearance is an expected feature, not an edge case, of the CO_2 extension. In physical wells, heat ingress from the completion and formation partially offsets this cooling, which is the motivation for the conjugate thermal model of the next phase; a prescribed-wall-temperature benchmark (Dittus–Boelter coupling, $T_w = 300$ K) indicates the magnitude of that offset.

7 Conjugate completion–formation heat transfer

7.1 Radial thermal network

The conjugate model attaches to every axial cell j a one-dimensional radial thermal network representing the concentric completion of the LSU CO_2 research well over its upper 2000 ft (table 3). Only the 7-5/8 in inner casing contains the pressurized fluid; the annulus between the 7-5/8 in and

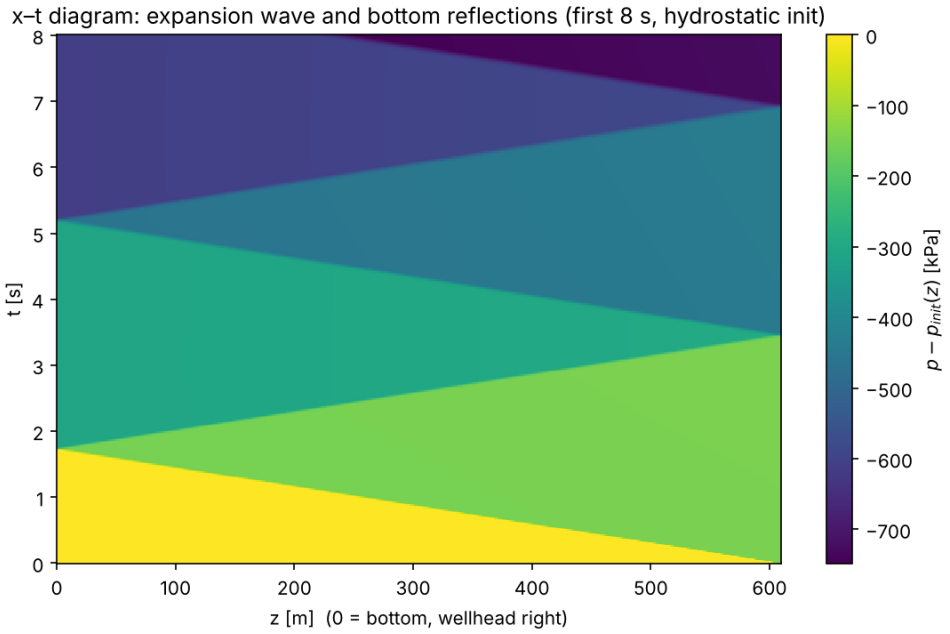


Figure 3: Distance–time (x–t) diagram of the pressure perturbation $p - p_{\text{init}}(z)$ during the first 8 s after valve opening (hydrostatic initialization): descending expansion wave, bottom reflection, and staircase reverberation of period $2L/a_0$.

13-3/8 in casings is air-filled, and the outer annuli are cemented to surface. The 30 in conductor (upper ∼150 ft only) is neglected.

Table 3: Radial stack (upper 2000 ft; radii from the casing table and API wall thicknesses: 7-5/8 in 35.8 lb/ft, 13-3/8 in 72 lb/ft, 20 in 133 lb/ft; 26 in borehole). All properties are parameters.

Region	$r_{\text{in}} \to r_{\text{out}}$ [mm]	Model	k [W/(m K)]
Fluid bore	$0 \to 85.2$	1-D flow; Dittus–Boelter h	—
7-5/8 in CRA casing	$85.2 \to 96.8$	capacitive shell	19
Air annulus	$96.8 \to 156.8$	nonlinear conductance	Eq. (11)
13-3/8 in casing	$156.8 \to 169.9$	capacitive shell	45
Cement	$169.9 \to 237.9$	capacitive shells	0.9
20 in casing	$237.9 \to 254.0$	capacitive shell	45
Cement	$254.0 \to 330.2$	capacitive shells	0.9
Formation	$330.2 \to 5000$	16 log-spaced shells	2.1

Each solid shell i carries the capacitance per unit length $C'_i = \rho_i c_{p,i}\,\pi\,(r^2_{\text{out},i} - r^2_{\text{in},i})$, lumped at the logarithmic-mean radius, and adjacent shells are connected by cylindrical-shell conduction resistances $R' = \ln(r_b/r_a)/(2\pi k)$. The network solves

$$C'_i \frac{dT_i}{dt} = G'_{i-1/2}\,(T_{i-1} - T_i) + G'_{i+1/2}\,(T_{i+1} - T_i), \tag{10}$$

with the fluid coupling at the innermost node through the series conductance of the convective film and the inner half-shell, $G'_f = [\,1/(2\pi r_1 h) + R'_{\text{half}}\,]^{-1}$, h from Dittus–Boelter with a laminar floor

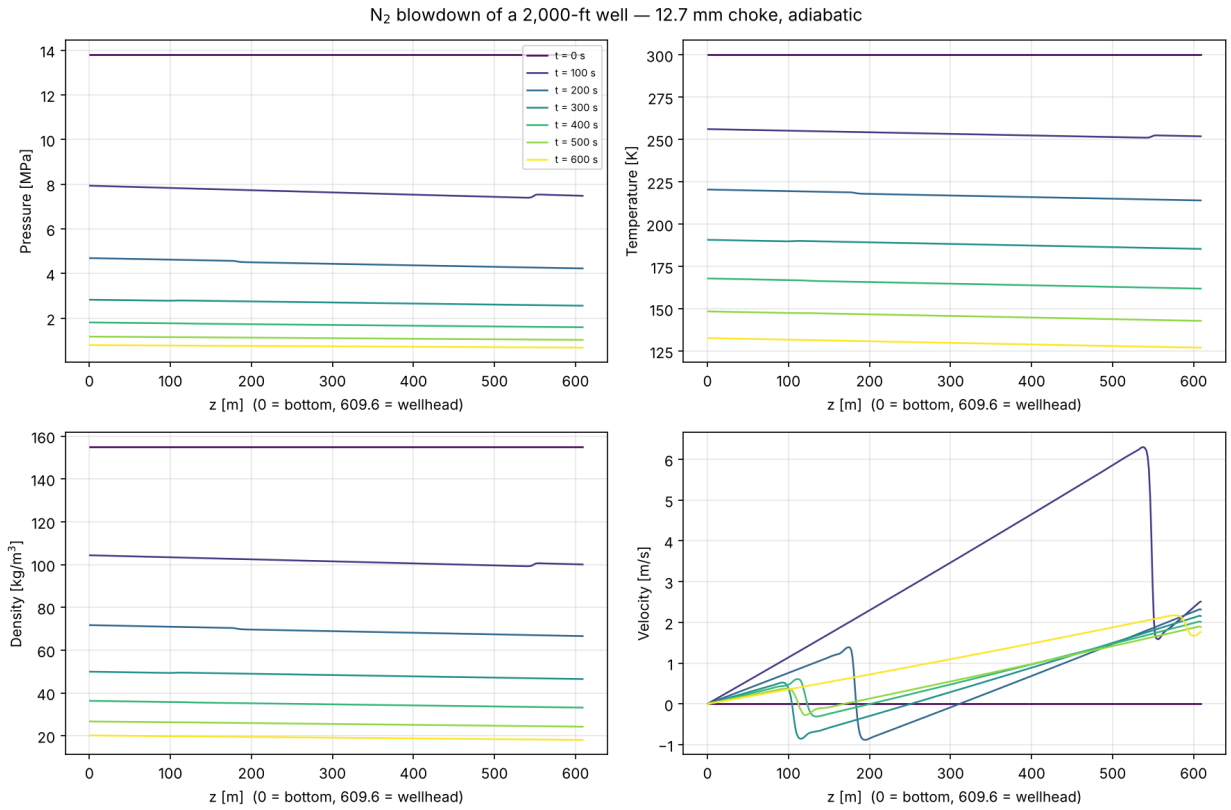


Figure 4: Axial profiles of pressure, temperature, density, and velocity at 100 s intervals during the baseline blowdown.

($\mathrm{Nu} \geq 3.66$), and the outer boundary held at the undisturbed geothermal temperature $T_{\mathrm{geo}}(z) = T_s + \Gamma\,(L - z)$ (defaults 294.3 K, 25.5 K/km). For blowdown time scales the rock penetration depth $\sqrt{\alpha t}$ is of order centimeters, so the far-field radius of 5 m is generous.

Air annulus. The gas-filled annulus is treated as a massless nonlinear conductance following the standard wellbore treatment of Willhite (1967): natural convection through an effective conductivity from the Dropkin–Somerscales correlation,

$$k_{\mathrm{eff}} = \max\Big[\, k_{\mathrm{air}},\; 0.049\,(\mathrm{Gr}\,\mathrm{Pr})^{1/3}\,\mathrm{Pr}^{0.074}\,k_{\mathrm{air}} \Big], \qquad \mathrm{Gr} = \frac{g\,\beta\,\Delta T\,(r_o - r_i)^3}{\nu^2}, \tag{11}$$

in parallel with gray-surface radiation between concentric cylinders ($\varepsilon = 0.8$ on both surfaces, linearized conductance $2\pi r_i\,\sigma F\,(T_i^2 + T_o^2)(T_i + T_o)$ with the standard two-surface view factor F). The conductance is re-evaluated every time step from the current surface temperatures (coefficient lagging), which keeps the implicit solve linear.

7.2 Solution procedure and coupling

The networks are advanced once per flow time step with backward Euler; each axial cell requires one tridiagonal (Thomas) solve, vectorized across cells. The fluid temperature enters the solid solve

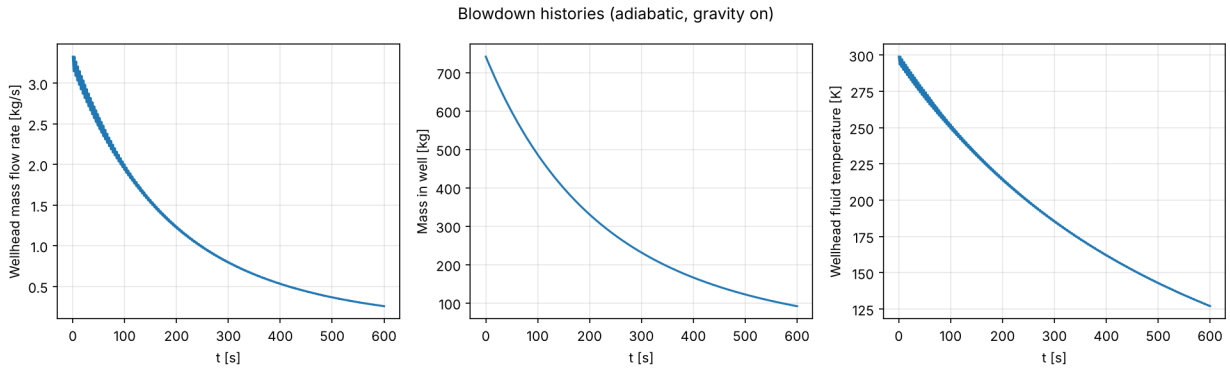


Figure 5: Wellhead mass flow rate, mass inventory, and wellhead fluid temperature during the baseline blowdown.

explicitly, while the heat flux returned to the wellbore energy equation,

$$q''_w = \frac{G'_f \left(T_1^{n+1} - T_f\right)}{2\pi r_1}, \tag{12}$$

uses the *implicit* wall temperature through the same conductance. The energy delivered by the fluid source term in eq. (3) is therefore exactly the energy absorbed by the network, and the global budget of eq. (7), extended by the wall-heat integral $\int \sum_j q''_w \pi D \, \Delta z \, dt$, continues to close to machine precision in the conjugate case.

7.3 Verification

Four tests establish the thermal model (all automated in the regression suite):

1. *Steady state:* with constant fluid and far-field temperatures, the converged heat flow is identical across every radial interface (series-resistance consistency) to 10^{-6} relative.

2. *Energy conservation:* with an insulated outer boundary, the heat absorbed from the fluid equals the change of stored network energy to 10^{-10} (backward Euler is exactly conservative).

3. *Transient conduction:* a formation-only network driven by a constant wall temperature reproduces the constant-temperature-cylinder solution to better than 1 % over $\tau = \alpha t / r_w^2 \in [10, 1000]$, where the reference is an independent Crank–Nicolson solution on a 6000-node logarithmic-radial grid. A methodological finding is recorded here explicitly: three convenient references were tried first and all proved inadequate — the leading-order logarithmic asymptote is 8–15 % in error over this τ range; Ramey-type dimensionless-temperature correlations (Ramey, 1962; Hasan and Kabir, 1991) address the constant-*flux* boundary condition rather than constant temperature; and naive quadrature of the exact Carslaw–Jaeger integral (Carslaw and Jaeger, 1959) mis-handles the singular $u \to 0$ region of the integrand that dominates at large τ. The network initially appeared 3–7 % "in error" against these references; the independent finite-difference solution showed the network to be accurate to 0.3 % and the references to be at fault.

4. *Radial grid convergence* of $q''_w(t)$ under refinement of the formation discretization (deviation < 2 % at 16 shells against 32).

7.4 Results: thermal-model hierarchy

Figure 6 compares the three thermal levels on the actual well geometry (bore ID 170.3 mm, 12.7 mm choke, otherwise the baseline case) over 1800 s. The outcome is not a small correction: heat ingress from the completion and formation nearly suppresses the expansion cooling. The minimum wellhead fluid temperature is 283.6 K for the coupled model and 290.2 K for the prescribed-T_w benchmark, against 124.0 K for the adiabatic wall. The warmer inventory sustains a higher pressure at equal mass, so the heated cases also discharge measurably faster.

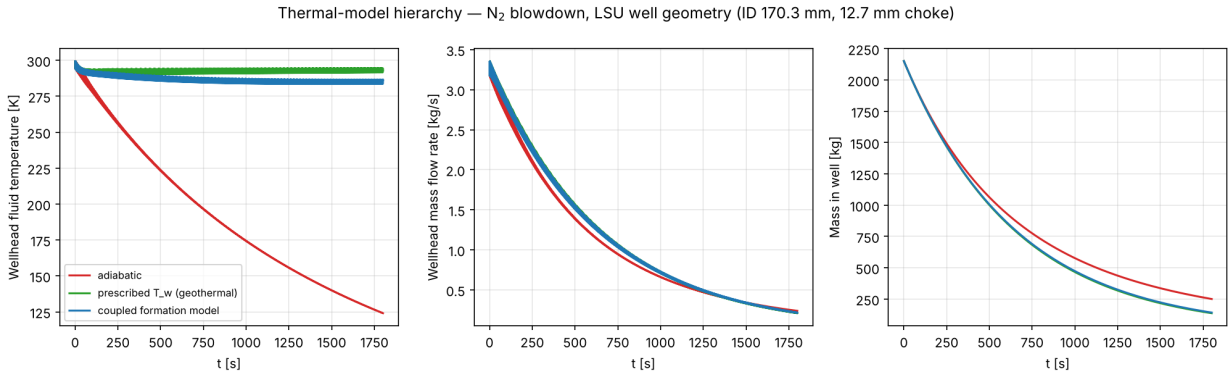


Figure 6: Thermal-model hierarchy on the LSU well geometry (N_2, 13.79 MPa/300 K, 12.7 mm choke): wellhead fluid temperature, mass flow rate, and inventory for the adiabatic, prescribed-T_w, and coupled formulations.

Figure 7 locates the controlling resistances: of the total fluid-to-formation temperature difference, roughly 3 K falls across the convective film, 7 K across the air annulus, and less than 1 K across all remaining solids; the formation beyond the cement is essentially undisturbed after 30 min. Two consequences follow. First, the air-annulus closure — here the Willhite/Dropkin–Somerscales treatment, developed for modest gap aspect ratios — is a leading-order sensitivity of the model and merits a dedicated assessment if annulus conditions (fluid, pressure, emissivities) change. Second, the near-isothermal outcome is specific to this slow (∼10 min) discharge; larger chokes shorten the residence time relative to the thermal response and shift the behavior back toward the adiabatic limit.

Limitations. The bore is modeled as a clean circle at the 7-5/8 in ID; the 2-7/8 in tubing, injection line, and gauge carriers inside the casing are not yet represented hydraulically or thermally. Axial conduction in the solids is neglected. Fluid-side transport properties are constant (to be revisited for transcritical CO_2).

8 Conclusions and outlook

A conservative Godunov/HLLC finite-volume model of vertical-well blowdown has been implemented and verified for the adiabatic, ideal-gas stage: discrete mass and total-plus-potential energy budgets close to machine precision; the quasi-static limit reproduces the lumped-vessel solution to better than 1 %; and the characteristic-coupled choke boundary condition behaves consistently from the choked regime through valve closure, including the open-pipe limit. The baseline nitrogen case exhibits the expected physics: a weak descending expansion wave and reverberation, quasi-static

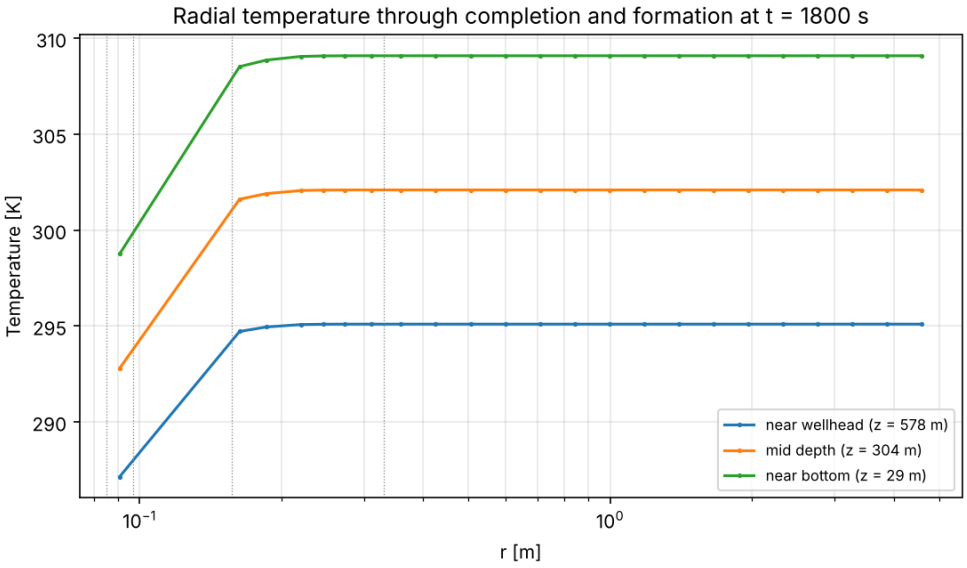


Figure 7: Radial temperature through the completion and formation at $t = 1800\,\mathrm{s}$ at three depths (coupled model). The dominant solid-side temperature drop occurs across the air annulus.

discharge, and strong expansion cooling that reaches the nitrogen critical temperature at the end of the transient.

The conjugate completion–formation thermal model closes the preferred level of the thermal hierarchy: an implicitly advanced radial network of the actual LSU well completion, verified against analytical limits and an independent transient-conduction reference, coupled to the wellbore energy equation with exact discrete energy exchange. Its first application overturns the adiabatic intuition for this facility: for the baseline slow blowdown, heat supplied by the casing, cement, and near-well formation holds the wellhead fluid within $\sim$16 K of the geothermal state (minimum 284 K against 124 K adiabatic), with the air-filled 7-5/8 in $\times$ 13-3/8 in annulus as the dominant solid-side resistance. The practical corollary is that the severity of blowdown cooling at this facility is controlled by the choke size (residence time) and by the annulus condition — both operationally accessible parameters.

The next stages, in order of priority: (i) real-fluid CO_2 thermodynamics behind the existing property interface (single-phase gaseous, then dense/transcritical), where the competition between Joule–Thomson cooling and formation heat ingress — now resolvable with the verified conjugate model — is the physically interesting question for facility blowdown tests; and (ii) phase appearance and two-phase CO_2 flow, building on the two-phase benchmark heritage of the framework.

Code and data availability

The implementation, verification scripts, and notebooks are maintained at https://github.com/JaderBarbosaLSU/hyperco2-wellbore.

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